

SDS 5380 Social Algorithms Project Proposal

Group Members: Hua Deng, Grace Yin, Hayley Shin, Tracy Chen, Gerald Velasquez

Paper: Shifting Attention to Accuracy Can Reduce Misinformation Online” (Pennycook et al., 2021)

Paper Link: <https://www.nature.com/articles/s41586-021-03344-2>

This project aims to replicate the paper “Shifting Attention to Accuracy Can Reduce Misinformation Online” (Pennycook et al., 2021), which examines why individuals share misinformation on social media and proposes a simple intervention to reduce it. The paper argues that misinformation sharing is driven primarily by inattention to accuracy rather than confusion or deliberate preference for false content. Through a series of survey experiments (Studies 1-6) and a field experiment (Study 7), the authors show that prompting individuals to consider accuracy can improve the quality of the content they choose to share.

For our replication, we focus on Studies 1-6, for which the authors provide datasets and Stata code. However, the provided Stata code does not run as-is due to non-standard commands (e.g., `cluster2`, `logit2`). These appear to be custom or unavailable packages, and required `.ado` files are not included in the shared materials. Thus, given these limitations, we treat the Stata code as a reference for model specification and plan to re-implement the analyses in Python, ensuring reproducibility while closely following the original empirical design. In addition, Study 7 uses Twitter data that are not publicly available, so we focus on replicating Studies 1-6.

Overall, this project follows a Template 2 approach, focusing primarily on replicating an analysis from a published paper, possibly adding a small extension. In the following section, we describe what we aim to replicate and possible extensions for each study.

Study 1 randomly assigned participants to either judge the accuracy of 36 true and false headlines or indicate whether they would share them on social media (between-subjects design). We plan to replicate the full analysis to: (1) validate reported statistics such as the 55.9 pp veracity effect in the accuracy condition and the 5.9 pp veracity effect in the sharing condition, and (2) identify limitations including reliance on self-reported sharing intentions and a sample restricted to U.S. MTurk workers evaluating only political headlines.

Study 2 replicates the finding that participants rated accuracy as significantly more important than all other sharing motivations, including partisan alignment, surprise, interestingness, and humor (all $p < 0.001$). We add effect sizes omitted from the original paper ($d = 0.94$ vs. surprise, $d = 0.69$ vs. politics, $d = 0.32$ vs. interestingness) and Wilcoxon robustness checks confirming results hold under ordinal assumptions.

Studies 3-5 test whether subtly priming accuracy improves people's ability to distinguish between true and false news when deciding what to share. We plan to replicate the main analyses (condition $\times$ veracity effects) using the provided code and verify that results are consistent with the paper. The main analysis conditions on a post-treatment variable, which is partially addressed by chi-square tests reported in the paper (but not implemented in the code) and by robustness analyses; we will implement these chi-square tests to complete the replication. As a possible extension, we examine whether the treatment effect differs by cognitive reflection ability (CRT scores), which were collected in Studies 3-5 but never analyzed. If the condition $\times$ veracity $\times$ CRT interaction is not significant, this provides novel evidence that accuracy priming works equally well across individuals with different levels of reflective thinking, strengthening the paper's claim that the intervention is broadly scalable.

Study 6 is a survey experiment that quantifies why people share misinformation by decomposing false-news sharing into three causes: inattention (failing to consider accuracy), confusion (mistakenly believing false content is true), and purposeful sharing (knowingly sharing falsehoods). Our replication will focus on the core decomposition analysis reported in Figure 3d — estimating the share of false-news sharing attributable to inattention, confusion, and purposeful sharing — using the provided raw data. We will additionally attempt to reproduce the bootstrap confidence intervals, which were reported in the paper but absent from the shared code.